Pengfei Hu

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Education

Stevens Institute of Technology Sep. 2023 – Present
Ph.D. in Computer Science (GPA: 3.95/4.00) *Hoboken, NJ*

- **Ph.D. Advisor:** Dr. Yue Ning
- **Research Topic:** Knowledge-Guided, Robust and Trustworthy Clinical AI;
- **Specific Keywords:** Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Transfer Learning / Post-Training (Domain Adaptation), Knowledge Graph (KG), Large Language Model (LLM), LLM Agent, Graph Neural Network (GNN), Temporal Event Prediction, Explainable AI (XAI).

University of Southern California Sep. 2021 – May 2023
M.S. in Industrial System Engineering (GPA: 4/4) *Los Angeles, CA*

- Research Assistant advised by Dr. Sze-chuan Suen working on monitoring of diabetes via wearable devices.

Work Experience

Oak Ridge National Laboratory May 2025 – Aug. 2025
Research Scientist Intern, Mentor: Ming Fan, Leader: Dan Lu *Oak Ridge, TN*

- Utilized the Frontier supercomputer to deploy parallel training on over 10 compute nodes via DDP and mpi4py;
- **Project 1:** Develop adaptive graph learning with Transformer for inflow prediction in Upper Colorado Reservoirs;
- **Project 2:** Explore feature modulation with meta attributes for enhancing model generalization across reservoirs;
- **Paper:** One journal paper in Environmental Modelling & Software and one AAAI workshop (oral) paper.

Research Experience

Interpretable Clinical Domain Adaptation | *Keywords: Sparse Autoencoder, XAI* Dec. 2025 – Present

- Designed an orthogonal inference module to decompose invariant label signals and domain-specific covariates upon concept-grounded latent geometry, improving robustness under spatial and temporal distribution shifts;
- Provided transparent cross-domain evidence through sparse vector ablation and medical concept attribution;
- **Paper:** Currently under review, available on arXiv preprint 2602.12542.

Knowledge-Guided Diagnostic Reasoning | *Keywords: LLM, KG, GNN* Apr. 2025 – Feb. 2026

- Built an LLM-enhanced (Llama 3.1-8B) medical knowledge graph integrating 9 clinical knowledge sources (e.g., HPO, DrugBank), and designed a self-supervised pretraining task to model stepwise lab-informed reasoning;
- Developed a dual-expertise model combining KG embeddings and GNN for interpretable and accurate predictions;
- **Paper:** Published in IEEE Journal of Biomedical and Health Informatics (IF: 6.8, March 2026).

LLM Agents for Causal Discovery | *Keywords: AI Agent, RAG, LangChain* Feb. 2025 – Nov. 2025

- Introduced a multi-agent LLM (ChatGPT4o-mini) framework with Knowledge Retrieval, Causal Discovery (generate causal graphs), and Decision-Making agents for interactable and interpretable healthcare prediction;
- Utilized RAG with a vector database (ChromaDB with Sentence-BERT) to support agent reasoning;
- **Paper:** Published in Findings of EMNLP 2025 (Nov 2025).

Selected Publications

1. **Pengfei Hu**, Chang Lu, Fei Wang, and Yue Ning, "Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning", Accepted by *IEEE Journal of Biomedical and Health Informatics (IF: 6.8)*, 2026.
2. Xiaoxue Han, **Pengfei Hu**, Jun-En Ding, Chang Lu, Feng Liu, Yue Ning, "Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery", In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025.
3. **Pengfei Hu***, Eric Yang*, Xiaoxue Han, and Yue Ning, "MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records", In *2024 IEEE International Conference on Big Data (IEEE BigData)*, 2024.

Technical Skills

Programming Languages: Python, SQL (Postgres), Java, R, Latex.
Deep Learning Frameworks & Libraries: Pytorch, Keras, PyG, pandas, NumPy, Matplotlib.